

Playwright Capstone Project Documentation

Project Title

Playwright Capstone Project

Project Type

Enterprise Quality Engineering Automation Capstone Project

Developed During

Playwright Automation Training Program

Project Objective

The objective of this capstone project is to design and implement a scalable automation testing framework using Playwright.

The project focuses on automating major functionalities of a web application while following real-world QA automation practices such as:

- End-to-End Testing
 - Functional Testing
 - UI Validation
 - Reusable Framework Design
 - Cross Browser Testing
 - Automation Reporting
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Project Website

Primary Website Used: Razer

Website URL: <https://www.razer.com/>

Tools & Technologies Used

Category	Technology
Automation Framework	Playwright
Programming Language	JavaScript
Runtime Environment	Node.js
IDE	Visual Studio Code
Version Control	Git & GitHub
Reporting Tool	Playwright HTML Report
Browser Support	Chromium, Firefox, WebKit

Modules Covered

1. Authentication Module

- User Login
- User Logout
- Invalid Login Validation

2. Registration Module

- New User Registration
- Mandatory Field Validation

3. Product Module

- Product Search
- Product Details Validation

4. Cart Module

- Add to Cart
- Remove from Cart

5. Wishlist Module

- Add to Wishlist
- Remove from Wishlist

6. Checkout Module

- Billing Validation
- Shipping Validation

7. Payment Module

- Payment Method Selection
- Order Confirmation Validation

8. User Profile / Order History Module

- User Information Validation
 - Order History Validation
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Automation Framework Features

- Playwright Test Runner
 - Page Object Model (POM)
 - Reusable Test Methods
 - Modular Test Design
 - Parallel Test Execution
 - Assertions & Validations
 - Screenshots on Failure
 - Trace Viewer Support
 - Headed & Headless Execution
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Folder Structure

playwright-capstone-project/

```
├── tests/
├── pages/
├── utils/
├── test-data/
├── screenshots/
├── playwright.config.js
├── package.json
└── README.md
```

Test Execution Commands

Run All Tests

```
npx playwright test
```

Run Tests in Headed Mode

```
npx playwright test --headed
```

Run Tests on Chromium Browser

```
npx playwright test --project=chromium
```

Open Report

```
npx playwright show-report
```

Conclusion

This capstone project demonstrates practical understanding of automation testing concepts using Playwright.

The project focuses on building hands-on experience in:

- Enterprise QA Workflows
- Automation Framework Development
- Test Case Automation
- Browser Testing
- End-to-End Validation
- Scalable Test Architecture